Project Report: Online Banking Application

Course: DBMS Project (Database Management System)

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1. Introduction

Online Banking Applications allow customers to perform banking transactions and access banking services through the internet. This system provides a secure platform for managing accounts, transferring funds, paying bills, and monitoring transactions, reducing the dependency on physical bank branches.

This project simulates a simplified version of an online banking system with user authentication, account management, and transaction handling, implemented using a database to manage user and transaction information.

2. Objectives

- To provide a secure, efficient, and user-friendly online banking system.
- To enable customers to perform banking operations online, such as deposits, withdrawals, and fund transfers.
- To maintain an organized and consistent database of users, accounts, and transactions.
- To implement access control for users via login, registration, and session management.
- To give administrators tools to manage accounts and monitor transactions.

3. Technology Stack

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP
- **Database:** MySQL
- **Server:** Apache (XAMPP/LAMP/WAMP)

4. Database Schema

4.1 Tables

1. **Users**

- `user_id` (INT, PK)
- `name` (VARCHAR)
- `email` (VARCHAR, UNIQUE)
- `password` (VARCHAR)
- `phone` (VARCHAR)
- `account_id` (INT, FK → Accounts.account_id)

2. **Accounts**
   - `account_id` (INT, PK)
   - `account_type` (VARCHAR)
   - `balance` (DECIMAL)
   - `created_at` (DATETIME)
3. **Transactions**
   - `transaction_id` (INT, PK)
   - `account_id` (INT, FK → Accounts.account_id)
   - `transaction_type` (VARCHAR) [Deposit/Withdrawal/Transfer]
   - `amount` (DECIMAL)
   - `transaction_date` (DATETIME)

5. Core Features & Functionality

- **User Registration:** Users can register with personal details. Passwords are encrypted in the database.

- **Login/Logout:** Secure login with session management. Users can log out to terminate sessions.
- **Account Management:** Users can view account details, check balance, and update contact information.
- **Transactions:** Users can perform deposits, withdrawals, and fund transfers between accounts.
- **Transaction History:** Users can view past transactions in detail.

6. User Authentication (Session Management)

- **Registration:** Users fill a form with name, email, password, and phone number. A new account is created in the database.
- **Login:** Users provide email and password. Sessions are initiated upon successful login.
- **Logout:** Users can log out to destroy the session and prevent unauthorized access.

7. Main Dashboard (`dashboard.php`)

The main dashboard serves as the central interface for users. It provides:

1. **Create Account:** Users can create multiple accounts under their profile.
2. **Read/View:** Users can view account details, balances, and transaction history.
3. **Update:** Users can update contact information or account details (if allowed).
4. **Delete:** Users can request account deletion, which removes their account from the database.

Example functionality workflow:

- **View Account:** Query the `Accounts` table filtered by `user_id`.
- **Transfer Funds:** Insert transaction records in `Transactions` table and update `Accounts.balance`.
- **Transaction History:** Fetch records from `Transactions` table based on `account_id`.

8. Conclusion

The Online Banking Application provides a secure, reliable, and convenient platform for banking operations. By implementing a robust database management system, it ensures data integrity and enables efficient handling of user accounts and transactions. This project demonstrates core DBMS concepts such as table relationships, CRUD operations, and transaction management in a real-world scenario.